Presentation: Monotone maps, sphericity and bounded second eigenvalue by Bilu, Linial

CSE 269

Ian Chi

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Key results

- $\boxed{\begin{array}{l}\delta n \text{ regular graphs with bounded diameter D have sphericity } \Omega(\frac{n}{\lambda_2+1}) \\ \text{where } \lambda_2 \text{ is the second largest eigenvalue of the adjacency matrix}\end{array}}$
- quasi-random graphs have logarithmic sphericity
- $\frac{n}{2}$ regular graphs w/ bounded λ_2 are $o(n^2)$ close to bipartite graphs

Sphericity

Definition (sphericity)

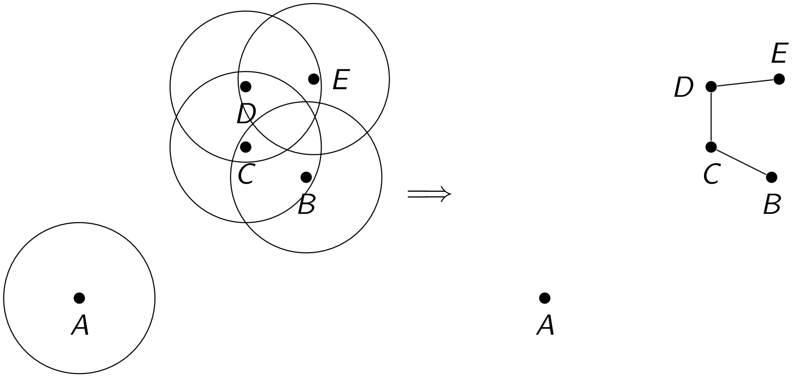
given a graph G and adjacency matrix A, its sphericity is the lowest dimension n ST $\varpi : G - \mathbb{R}^n$ satisfies

$$A_{ij} = 1 \iff |\varphi(i) - \varphi(j)|_2 \leq 1 \tag{1}$$

- this map would be an isometry if we used the metric on the original graph of

$$\begin{aligned} \delta'(i,j) &= 0 \iff i = j \\ \delta'(i,j) &= 1 \iff \delta(i,j) = 1 \\ \delta'(i,j) &= 2, \text{ else} \end{aligned} \tag{2}$$

Sphericity



(3)

spectral graph theory

- use linear algebra/spectral theory on adjacency matricies for info
- n regular graphs have largest eigenvalue of n
- second eigenvalue gives good info on "variance" of edges by expander mixing lemma for some subset $S, T \subset G$

$$\left| E(S,T) - \frac{d}{n}|S||T| \right| \leq \lambda_2 \sqrt{|S||T|} \tag{4}$$

- λ_2 controls for "worst case deviation"
- good expander graphs lack local clusters; local clusters are easy to compress

Theorem Statement

Theorem

Let G be a d-regular graph with diameter D and λ_2 be the second largest eigenvalue of G's adjacency matrix and at least $d - \frac{1}{2}n$. Then $Sph(G) = \Omega\left(\frac{d-\lambda_2}{D^2(\lambda_2+O(1))}\right) = \Omega(\frac{n}{\lambda_2+1})$.

proof

a couple observations:

- if X is real symmetric,

$$rank(X) \geq \frac{(tr(x))^2}{\sum_{i,j} X_{ij}^2} \tag{5}$$

- if X is PSD, then $\exists$ a set v_i such that $X_{ij} = v_i \cdot v_j$. X is also called a Grammian matrix
- if the embedding of an enumerated graph $w_i \in V$ is given by

$$\varphi : w_i \mapsto v_i \tag{6}$$

  for vectors v_i comprising of the grammian matrix X, then the dimension of the embedding $= rank(X)$

Idea: what we will do is construct a feasable grammian matrix where the rank is $\frac{d-\lambda_2}{D^2(\lambda_2+O(1))} \approx \frac{n}{\lambda_2+1}$

proof (cont.)

- X: nxn symmetric matrix
- J: all ones matrix
-
$$R : (i,j) \in \mathbb{R}^{n\times n} \mapsto X_{ii} \tag{7}$$
$$R(X) \in \mathbb{R}^{n\times n}$$
rows are all the same element, ith row is just all X_{ii}
- Y: $2X - R(X) - R(X)^T + J$

- why is Y like that?
  - $Y_{ii} = 1$
  - $$Y_{ij} = 2\langle x_i, x_j\rangle - \langle x_i, x_i\rangle - \langle x_j, x_j\rangle + 1 \\ = 1 - |x_i - x_j|^2 \tag{8}$$
- applying rank inequality 5

$$rank(Y) \geq \frac{n^2}{n + \sum_{i \neq j} \left(1 - |x_i - x_j|^2\right)^2} \tag{9}$$

- new goal: show

$$\sum_{i \neq j} \left(1 - |x_i - x_j|^2\right)^2 = O(D^2 n^2 \frac{\lambda_2}{d - \lambda_2}) \tag{10}$$

proof (cont.)

- we can normalize by D^2. now all $|x_i - x_j| \approx 1$.
- separate ij pairs into two groups
  - ij is an edge,
  - ij is not an edge

  feasible embedding $\implies$

$$\begin{cases} |v_i - v_j| \leq 1 & \iff (i,j) \in E(G) \\ |v_i - v_j| > 1 & \iff (i,j) \notin E(G) \end{cases} \tag{11}$$

- max eq 10 $\iff$

$$\max \sum_{ij \notin E} \left(|x_i - x_j|^2 - 1\right) + \sum_{ij \in E} \left(1 - |x_i - x_j|^2\right) \tag{12}$$

proof (cont.)

$$\begin{array}{rl}
\max & \sum_{(i,j)\notin E}(V_{ii}+V_{jj}-2V_{ij}) + \sum_{(i,j)\in E}(-V_{ii}-V_{jj}+2V_{ij}) - \binom{n}{2} + nd \\
s.t. & V \in PSD \\
\forall (i,j)\in E & V_{ii}+V_{jj}-2V_{ij} \le 1 \\
\forall (i,j)\notin E & V_{ii}+V_{jj}-2V_{ij} \ge 1
\end{array}$$

The dual problem is:

$$\begin{array}{rl}
\min & \frac{1}{2} trA \\
s.t. & A \in PSD \\
\forall (i,j)\in E & A_{ij} \le -1 \\
\forall (i,j)\notin E, i \neq j & A_{ij} \ge 1 \\
\forall i \in [n] & \sum_{j=1,\ldots,n} A_{ij} = 0
\end{array}$$

proof (cont.)

Create matrix that satisfies conditions for A

- the matrix $A = (\alpha d - n)I + J - \alpha M$ where M is the adjacency matrix satisfies all of these conditions if $\alpha > 2$.
- on edges, $A_{ij} = 1 - \alpha$, on non edges $A_{ij} = 1$ and we want edge A_{ij} to be less than -1 by our optimization problems, thus $\alpha \geq 2$
- finally, we can check that A's eigenvalues are the same as Ms, with an eigenvector u having eigenvalue $(\alpha d - n) - \alpha \lambda_i$. to ensure this is psd, alpha is set to $\frac{n}{d - \lambda_2}$.
  - note that $A\vec{1} = 0$
- This A gives an upper bound on the value 12

proof (cont.)

$$\begin{aligned}\frac{1}{2}trA =& \frac{1}{2}n(\alpha d - n + 1)\\ =& \frac{1}{2}\left(n^2\frac{d}{d-\lambda_2} - n^2 + n\right)\\ =& \frac{1}{2}\left(n^2\frac{\lambda_2}{d-\lambda_2} + n\right)\end{aligned} \tag{13}$$

we have finished what we set out to prove in eq 10 □

some results on bipartite graphs

$$Sph(K_{m,n}) \leq m + \frac{n}{2} - 1 \tag{14}$$

$$Sph(K_{n,n}) \geq n \tag{15}$$